

Isentropic Efficiency of Adiabatic Fan and Compressor

$$(\cancel{\dot{Q}_{IN}} - \cancel{\dot{Q}_{OUT}}) + (\dot{W}_{IN} - \dot{W}_{OUT}) = \dot{m} (h_{0,OUT} - h_{0,IN})$$

(annotation on the crossed-out term: Adiabatic)

$$\dot{W}_{IN} = \dot{W}_{shaft} = \int \rho \dot{w}_{shaft} dV$$

$$\rho \dot{w}_{shaft} = \rho \vec{u} \cdot \nabla w_{shaft} = \rho \vec{u} \cdot \vec{f}_{shaft} \quad (\text{per unit volume})$$

This because viscous work opposes the “direction” of work which is transferring outwards

$$\dot{W}_{OUT} = -\dot{W}_{shear}; \quad \dot{W}_{shear} = \int (\tau \vec{u}) \cdot \hat{n} dS \quad \text{where } \hat{n} \text{ points OUTWARDS of control surface.}$$

$$\therefore \dot{W}_{shear} = \int \nabla \cdot (\tau \vec{u}) dV$$

$$\dot{W}_{shaft} + \dot{W}_{shear} = \dot{m} (h_{0,out} - h_{0,IN})$$

$$\dot{W}_{shaft} = \dot{m} (h_{0,OUT,s} - h_{0,IN})$$

$$(1) \quad \eta_s = \frac{h_{0,OUT,s} - h_{0,IN}}{h_{0,OUT} - h_{0,IN}} = \frac{\dot{m} (h_{0,OUT,s} - h_{0,IN})}{\dot{m} (h_{0,OUT} - h_{0,IN})} = \frac{\dot{W}_{shaft}}{\dot{W}_{shaft} + \dot{W}_{shear}}$$

$$\eta_s = \frac{1}{1 + \frac{\dot{W}_{shear}}{\dot{W}_{shaft}}} = \frac{1}{1 + \frac{\int \nabla \cdot (\tau \vec{u}) dV}{\int \rho \dot{w}_{shaft} dV}} = \frac{1}{1 + \frac{\int \nabla \cdot (\tau \vec{u}) dV}{\int \rho \vec{u} \cdot \vec{f}_{shaft} dV}}$$

Isentropic Efficiency of Adiabatic Compressor and Turbofan

$$\eta_s = \frac{1}{1 + \frac{\int \nabla \cdot (\tau \vec{u}) dV}{\dot{W}_{shaft}}}$$

$$\eta_s = \frac{1}{1 + \frac{\int \nabla \cdot (\tau \vec{u}) dV}{\int \rho \dot{w}_{shaft} dV}} \Rightarrow \eta_s(x, y) = \frac{1}{1 + \frac{\nabla \cdot (\tau \vec{u})}{\rho \dot{w}_{shaft}}}$$

$$\eta_s = \frac{1}{1 + \frac{\int \nabla \cdot (\tau \vec{u}) dV}{\int \rho \vec{u} \cdot \vec{f}_{shaft} dV}} \Rightarrow \eta_s(x, y) = \frac{1}{1 + \frac{\nabla \cdot (\tau \vec{u})}{\rho \vec{u} \cdot \vec{f}_{shaft}}}$$

Where $\dot{w}_{shaft}$ and $\vec{f}_{shaft}$ and $\dot{W}_{shaft}$ can be treated as constant over a determined control volume

Isentropic Efficiency of Adiabatic Compressor and Turbofan

$$\eta_s = \frac{\dot{W}_{shaft}}{\dot{W}_{shaft} + \dot{W}_{shear}} \quad \text{from equation (1)}$$

$$\eta_s = \frac{\int \rho \vec{u} \cdot \vec{f}_{shaft} dV}{\int \rho \vec{u} \cdot \vec{f}_{shaft} dV + \int \nabla \cdot (\tau \vec{u}) dV}$$

$$\nabla \cdot (\tau \vec{u}) = \vec{u} \cdot (\nabla \cdot \tau) + \tau : \nabla \vec{u}$$

Momentum N-S Equation

$$\rho (\vec{u} \cdot \nabla) \vec{u} = -\nabla P + \nabla \cdot \tau + \rho \vec{f}_{shaft}$$

$$\textcircled{1} \quad \rho (\vec{u} \cdot \nabla) \vec{u} + \nabla P = \nabla \cdot \tau + \rho \vec{f}_{shaft}$$

Now, P_0 is pressure at a spatial point where fluid has come to rest adiabatically and reversibly

$$(\underline{dq_{IN}} - \underline{dq_{OUT}}) + (dW_{IN} - \underline{dW_{OUT}}) = dh_0$$

(annotations: Adiabatic ; no irreversible viscous work)

Isentropic Efficiency of Adiabatic Compressor or Turbopan

$$dw_{in} = dw_{shaft}$$

$$\therefore dw_{shaft} = dh_0$$

$$dw_{shaft} = dh + u du; \quad dh_0 = d\left(h + \frac{1}{2}u^2\right)$$

$$\cancel{\mathcal{T}} d\cancel{s} = dh - \frac{dP}{\rho} = 0 \rightarrow dh = \frac{dP}{\rho} \quad (\text{annotation: } 0, \text{ isentropic})$$

$$dw_{shaft} = \frac{dP}{\rho} + u du$$

$$\frac{dw_{shaft}}{d(x,y)} = \frac{1}{\rho} \frac{dP}{d(x,y)} + \frac{u du}{d(x,y)}$$

$$\nabla w_{shaft} = \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u}$$

$$\vec{f}_{shaft} = \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u}$$

Isentropic Efficiency of Adiabatic Compressor or Turbofan

$\vec{f}(x, y, \vec{u}, t) = \vec{f}$, it is constant

$$\therefore \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u} - \vec{f}_{shaft} = 0$$

for all (x, y) isentropic

When $\vec{u} = 0$, $P = P_0$ and $\rho = \rho_0$

$$\therefore \frac{\nabla P_0}{\rho_0} + (\vec{0} \cdot \nabla) \vec{0} - \vec{f}_{shaft} = 0$$

$$\frac{\nabla P_0}{\rho_0} - \vec{f}_{shaft} = 0 = \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u} - \vec{f}_{shaft}$$

$$\therefore \frac{\nabla P_0}{\rho_0} = \vec{f}_{shaft} \quad \left(\textcircled{2} \frac{\nabla P_0}{\rho_0} - \cancel{\vec{f}_{shaft}} = \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u} - \cancel{\vec{f}_{shaft}} \right)$$

$$\textcircled{2} \quad \frac{\nabla P_0}{\rho_0} = \frac{\nabla P}{\rho} + (\vec{u} \cdot \nabla) \vec{u}$$

$$\therefore \textcircled{2} \quad \frac{\rho}{\rho_0} \nabla P_0 = \nabla P + \rho (\vec{u} \cdot \nabla) \vec{u}$$

Isentropic Efficiency of Adiabatic Compressor or Turbofan

$$\textcircled{1} \quad \rho (\vec{u} \cdot \nabla) \vec{u} + \nabla P = \nabla \cdot \tau + \rho \vec{f}_{shaft}$$

$$\textcircled{2} \quad \frac{\rho}{\rho_0} \nabla P_0 = \rho (\vec{u} \cdot \nabla) \vec{u} + \nabla P$$

$$\therefore \frac{\rho}{\rho_0} \nabla P_0 = \nabla \cdot \tau + \rho \vec{f}_{shaft}$$

$$\therefore \frac{\rho}{\rho_0} \nabla P_0 - \rho \vec{f}_{shaft} = \nabla \cdot \tau$$

Now

$$\nabla \cdot (\tau \vec{u}) = \vec{u} \cdot (\nabla \cdot \tau) + \tau : \nabla \vec{u}$$

$$\therefore \nabla \cdot (\tau \vec{u}) = \vec{u} \cdot \left(\frac{\rho}{\rho_0} \nabla P_0 - \rho \vec{f}_{shaft} \right) + \tau : \nabla \vec{u}$$

$$\therefore \nabla \cdot (\tau \vec{u}) = \frac{\rho}{\rho_0} \vec{u} \cdot \nabla P_0 - \rho \vec{u} \cdot \vec{f}_{shaft} + \tau : \nabla \vec{u}$$

Isentropic Efficiency of Adiabatic Compressor or Turbofan

Since,

$$\eta_s = \frac{\int \rho \vec{u} \cdot \vec{f}_{shaft} dV}{\int \rho \vec{u} \cdot \vec{f}_{shaft} dV + \int \nabla \cdot (\tau \vec{u}) dV}$$

$$\therefore \eta_s = \frac{\int \rho \vec{u} \cdot \vec{f}_{shaft} dV}{\int \rho \vec{u} \cdot \vec{f}_{shaft} dV + \int \left(\frac{\rho}{\rho_0} \vec{u} \cdot \nabla P_0 - \rho \vec{u} \cdot \vec{f}_{shaft} + \tau : \nabla \vec{u} \right) dV}$$

$$\eta_s = \frac{\int \rho \vec{u} \cdot \vec{f}_{shaft} dV}{\int \left(\frac{\rho}{\rho_0} \vec{u} \cdot \nabla P_0 + \tau : \nabla \vec{u} \right) dV}$$

or

$$\eta_s = \frac{\dot{W}_{shaft}}{\int \left(\frac{\rho}{\rho_0} \vec{u} \cdot \nabla P_0 + \tau : \nabla \vec{u} \right) dV}$$

or

$$\eta_s(x, y) = \frac{\rho \vec{u} \cdot \vec{f}_{shaft}}{\frac{\rho}{\rho_0} \vec{u} \cdot \nabla P_0 + \tau : \nabla \vec{u}}$$